

Fix loop declaration

Committed **before** the fix is written or benchmarked, so the prediction below is on record ahead of the result.

Date: 2026-10-01. Code at declaration: commit **HEAD** of this file's parent.

1. The worst-performing gate, with the failing number

G-REPEAT-ROOMS — the same flat walked twice does not produce the same number of rooms.

Measured by `bench/gates.py` over the two supplied whole-flat walks:

	1a8384c3f6	c7d28f72c6
Rooms found	3	6
Footprint	47.85 m ²	48.78 m ²

Two walks of one flat, differing by a factor of two in room count.

This is the worst *measured* gate. Ten other gates read NOT MEASURED — they need tape truth, or tiers that are not built — and a gate with no number behind it cannot be the subject of a fix loop, because there would be nothing to move.

Why this gate and not the footprint: G-REPEAT-FOOTPRINT is **MET**, at 1.9% between the same two walks. Floor area repeats; the division of that area into rooms does not. That pairing is the whole diagnosis in one line — the measurement is stable and the *segmentation* is not.

2. Root cause, and the evidence for it

Hypothesis. Room count is unstable because the seed step of the split admits cores far too small to be rooms, and how many such slivers survive depends on centimetre-scale differences in floor coverage that differ between any two walks.

`split_rooms` erodes the floor by half a doorway width, treats each surviving connected component as a room seed, then grows the seeds back under a watershed. The seed filter keeps any component of at least `max(min_area_cells // 4, 16)` cells — with a 1.20 m² minimum room area at 2 cm cells, that lower bound is **16 cells, or 0.0064 m²**. A patch of floor the size of a postcard becomes a room.

Evidence, measured on the two real walks before declaring (`bench/fix_loop_diagnosis.py`):

1. The surviving seed cores are mostly not rooms. At the shipped `door_max_m = 0.95`:

```
1a8384c3f6: 12 cores, areas 5.57 1.39 0.98 0.28 0.18 0.03 0.02 0.02 0.01 0.00 ...
c7d28f72c6: 9 cores, areas 8.70 2.09 0.78 0.71 0.40 0.10 0.02 0.01 0.00 ...
```

Three or four are plausible rooms. The rest are down to 0.00 m², and their number differs between the walks — 12 against 9.

2. The instability is a function of the one parameter the seeds depend on. Room count against erosion width, both walks:

<code>door_max_m</code>	1a8384c3f6	c7d28f72c6	
0.70	5	5	stable
0.80	5	5	stable
0.95 (shipped)	3	5	unstable
1.10	4	5	unstable
1.30	4	3	unstable
1.50	2	3	unstable

The two walks agree at 0.70 and 0.80 and disagree at every wider setting. This is a measurement on the real captures, not an argument from a synthetic case.

3. The underlying floor coverage is not the problem. 53.7 m² against 56.0 m², and 49.2 m² against 51.1 m² after walls are cut — about 4% apart, consistent with the 1.9% footprint agreement. The input to the split is stable; the split is not.

What this rules out. Drift is not the cause: corrections on these captures top out at 12 mm (`bench/results/gates.json`), far below the scale at which a 0.95 m erosion changes topology. Depth noise is

not the cause either: the coverage areas agree to 4%.

3. The fix to ship

1. **Filter seeds by a real minimum room area.** A seed core must be at least 0.25 m² — smaller than any room, far larger than a postcard — instead of the current 0.0064 m².
2. **Set the erosion width to the value the data says is stable:** `door_max_m` from 0.95 to 0.80, the widest setting at which both walks agree.

Both are parameter and filter changes inside `split_rooms`. Nothing else moves.

4. Prediction

Measured by `bench/gates.py` on the same two walks:

	Before	Predicted after
Rooms, 1a8384c3f6 vs c7d28f72c6	3 vs 6	5 vs 5
G-REPEAT-ROOMS	NOT MET	MET
G-REPEAT-FOOTPRINT	1.9% apart	1-4% apart, still MET
Footprint, 1a8384c3f6	47.85 m ²	46-50 m ²
Footprint, c7d28f72c6	48.78 m ²	47-51 m ²
A-RUNTIME	35.7 s worst	unchanged
Tests	27 pass	27 pass

Confidence: moderate-to-high on the room count, lower on the exact footprints. The `door_max_m` sweep is a direct measurement of the outcome being predicted — both walks give 5 at 0.80 today — so the headline number is close to read off. The seed-area filter is the part that is genuinely predicted rather than observed, and it may change the counts away from 5 in either direction.

Where this could still fall short. Agreeing on five rooms is not the same as being right about five rooms; the reference for these captures is nine. A narrower erosion also splits less aggressively, so genuinely separate rooms joined by a wide opening will merge. This fix targets *repeatability*, which is what the gate measures, and does not claim to fix room counts being low.

How to regenerate

```
git checkout <declaration commit> && python bench/gates.py # before
git checkout <fix commit> && python bench/gates.py # after
git diff <declaration commit> <fix commit> -- scanplan/ # the change
```